Lecture 24: The Cross Product

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

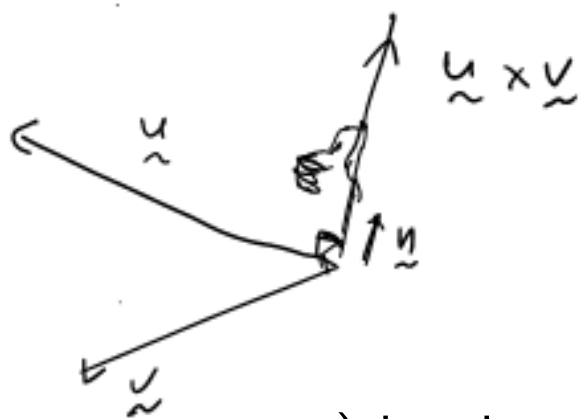
- ☐ Define the cross product of two vectors in three-dimensional space and describe its geometric significance.
- ☐ Compute the cross product using the determinant formula.
- ☐ Determine whether two nonzero vectors are parallel.
- ☐ Apply properties of the cross product in vector algebra.
- ☐ Calculate the area of a parallelogram and a triangle defined by two vectors using the magnitude of the cross product.
- ☐ Find a vector perpendicular to a given plane defined by two non-parallel vectors.
- ☐ Evaluate the scalar triple product and interpret it as the volume of a parallelepiped.
- ☐ Determine whether vectors are coplanar using the scalar triple product.

Cross Product

Let $\boldsymbol{u}$ and $\boldsymbol{v}$ be two non-parallel vectors in space, then they determine a plane.

We select a unit vector $\boldsymbol{n}$ perpendicular to the plane by the right-hand rule.





We define the cross product $\boldsymbol{u} \times \boldsymbol{v}$ (read as $\boldsymbol{u}$ cross $\boldsymbol{v}$) be the vector

$$\boldsymbol{u} \times \boldsymbol{v} = (|\boldsymbol{u}||\boldsymbol{v}| \sin\theta)\boldsymbol{n}$$

Note that $\boldsymbol{u} \times \boldsymbol{v}$ is parallel to the vector $\boldsymbol{n}$, so it is orthogonal to both $\boldsymbol{u}$ and $\boldsymbol{v}$.

Cross Product

Parallel Vectors

Nonzero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are parallel if and only if $\boldsymbol{u} \times \boldsymbol{v} = \boldsymbol{0}$.

$\underset{\sim}{u} \times \underset{\sim}{v} = (|\underset{\sim}{u}||\underset{\sim}{v}| \sin\theta)\,\underset{\sim}{n}$ $\quad$ $\underset{\sim}{u}$ and $\underset{\sim}{v}$ are parallel $\Rightarrow \theta = 0$ or π, $\sin\theta = 0$

Properties of the Cross Product

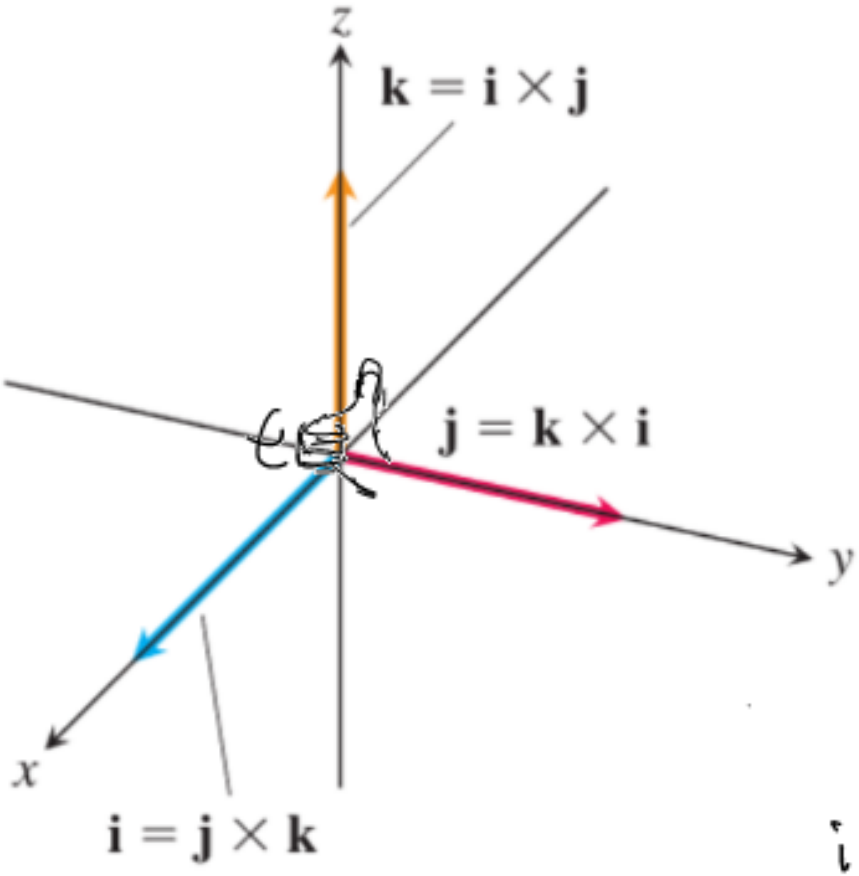
So, $\underset{\sim}{u} \times \underset{\sim}{v} = 0\,\underset{\sim}{n} = \underset{\sim}{0}$

If $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{w}$ are any vectors and r, s are scalars, then

1. $(r\boldsymbol{u}) \times (s\boldsymbol{v}) = (rs)(\boldsymbol{u} \times \boldsymbol{v})$
2. $\boldsymbol{u} \times (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \times \boldsymbol{v} + \boldsymbol{u} \times \boldsymbol{w}$
3. $\boldsymbol{v} \times \boldsymbol{u} = -(\boldsymbol{u} \times \boldsymbol{v})$ ← $\underset{\sim}{u} \times \underset{\sim}{v}$ and $\underset{\sim}{v} \times \underset{\sim}{u}$ are in opposite direction. Can be easily seen with Right hand rule.
4. $(\boldsymbol{v} + \boldsymbol{w}) \times \boldsymbol{u} = \boldsymbol{v} \times \boldsymbol{u} + \boldsymbol{w} \times \boldsymbol{u}$
5. $\boldsymbol{0} \times \boldsymbol{u} = \boldsymbol{0}$ If $\underset{\sim}{u} = \underset{\sim}{0}$ or $\underset{\sim}{v} = \underset{\sim}{0}$, then $\underset{\sim}{u} \times \underset{\sim}{v} = \underset{\sim}{0}$
6. $\boldsymbol{u} \times (\boldsymbol{v} \times \boldsymbol{w}) = (\boldsymbol{u} \cdot \boldsymbol{w})\boldsymbol{v} - (\boldsymbol{u} \cdot \boldsymbol{v})\boldsymbol{w}$

* In general, $\underset{\sim}{u} \times (\underset{\sim}{v} \times \underset{\sim}{w}) \neq (\underset{\sim}{u} \times \underset{\sim}{v}) \times \underset{\sim}{w}$ Not associative

Cross Product



By applying the right hand rule,

$\underset{\sim}{i} \times \underset{\sim}{j} = \underset{\sim}{k}$, $\underset{\sim}{j} \times \underset{\sim}{i} = -\underset{\sim}{k}$

$\underset{\sim}{j} \times \underset{\sim}{k} = \underset{\sim}{i}$, $\underset{\sim}{k} \times \underset{\sim}{j} = -\underset{\sim}{i}$

$\underset{\sim}{k} \times \underset{\sim}{i} = \underset{\sim}{j}$, $\underset{\sim}{i} \times \underset{\sim}{k} = -\underset{\sim}{j}$

Recall:

Two non zero vectors $\underset{\sim}{u}$ and $\underset{\sim}{v}$ are parallel iff $\underset{\sim}{u} \times \underset{\sim}{v} = \underset{\sim}{0}$

$\underset{\sim}{i} \times \underset{\sim}{i} = \underset{\sim}{j} \times \underset{\sim}{j} = \underset{\sim}{k} \times \underset{\sim}{k} = \underset{\sim}{0}$

$\underset{\sim}{i} \times (-\underset{\sim}{i}) = \underset{\sim}{j} \times (-\underset{\sim}{j}) = \underset{\sim}{k} \times (-\underset{\sim}{k}) = \underset{\sim}{0}$

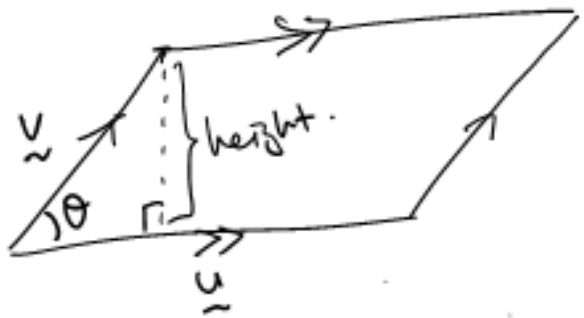
Area of a Parallelogram

Recall: $|\underset{\sim}{u} \times \underset{\sim}{v}| = |(|\underset{\sim}{u}||\underset{\sim}{v}| \sin\theta)\underset{\sim}{n}|$

$|\underset{\sim}{n}| = 1$

For $0 \le \theta \le \pi$, $0 \le \sin\theta \le 1$, $|\sin\theta| = \sin\theta$

length of $\underset{\sim}{u} \times \underset{\sim}{v}$ → $|\boldsymbol{u} \times \boldsymbol{v}| = |\boldsymbol{u}||\boldsymbol{v}| \sin\theta |\boldsymbol{n}| = |\boldsymbol{u}||\boldsymbol{v}| \sin\theta$



$\sin\theta = \frac{\text{height}}{|\underset{\sim}{v}|}$

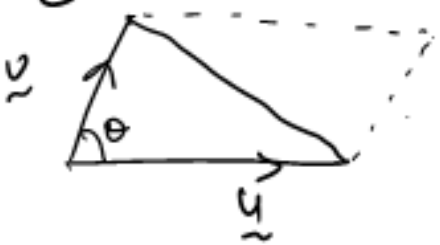
Area of parallelogram formed by $\underset{\sim}{u}$ and $\underset{\sim}{v}$

= base × height.

$= |\underset{\sim}{u}| \times |\underset{\sim}{v}| \sin\theta$

$= |\underset{\sim}{u} \times \underset{\sim}{v}|$

Similarly,



Area of triangle formed by $\underset{\sim}{u}$ and $\underset{\sim}{v}$

$= \frac{1}{2} \times$ Area of parallelogram formed by $\underset{\sim}{u}$ and $\underset{\sim}{v}$

$= \frac{1}{2}|\underset{\sim}{u} \times \underset{\sim}{v}|$

Determinant Formula for $\boldsymbol{u} \times \boldsymbol{v}$

Let $\boldsymbol{u} = u_1\boldsymbol{i} + u_2\boldsymbol{j} + u_3\boldsymbol{k}$ and $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$.

$$\boldsymbol{u} \times \boldsymbol{v} = (u_1\underset{\sim}{i} + u_2\underset{\sim}{j} + u_3\underset{\sim}{k}) \times (v_1\underset{\sim}{i} + v_2\underset{\sim}{j} + v_3\underset{\sim}{k})$$

$$= (u_1v_1)(\underset{\sim}{i}\times\underset{\sim}{i}) + (u_1v_2)(\underset{\sim}{i}\times\underset{\sim}{j}) + (u_1v_3)(\underset{\sim}{i}\times\underset{\sim}{k})$$
$$+ (u_2v_1)(\underset{\sim}{j}\times\underset{\sim}{i}) + (u_2v_2)(\underset{\sim}{j}\times\underset{\sim}{j}) + (u_2v_3)(\underset{\sim}{j}\times\underset{\sim}{k})$$
$$+ (u_3v_1)(\underset{\sim}{k}\times\underset{\sim}{i}) + (u_3v_2)(\underset{\sim}{k}\times\underset{\sim}{j}) + (u_3v_3)(\underset{\sim}{k}\times\underset{\sim}{k})$$

$$= (u_1v_2)\underset{\sim}{k} + (u_1v_3)(-\underset{\sim}{j}) + (u_2v_1)(-\underset{\sim}{k}) + (u_2v_3)\underset{\sim}{i} + (u_3v_1)\underset{\sim}{j} + (u_3v_2)(-\underset{\sim}{i})$$

$$= (u_2v_3 - u_3v_2)\underset{\sim}{i} - (u_1v_3 - u_3v_1)\underset{\sim}{j} + (u_1v_2 - u_2v_1)\underset{\sim}{k}$$

$$= \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}\underset{\sim}{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix}\underset{\sim}{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}\underset{\sim}{k}$$

determinant of matrix.

$$= \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

← determinant formula for cross product.

Calculating Cross Product

Example: Find $\boldsymbol{u} \times \boldsymbol{v}$ and $\boldsymbol{v} \times \boldsymbol{u}$ if $\boldsymbol{u} = 2\boldsymbol{i} + \boldsymbol{j} + \boldsymbol{k}$ and $\boldsymbol{v} = -4\boldsymbol{i} + 3\boldsymbol{j} + \boldsymbol{k}$.

$$\underset{\sim}{u} \times \underset{\sim}{v} = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} \underset{\sim}{i} - \begin{vmatrix} 2 & 1 \\ -4 & 1 \end{vmatrix} \underset{\sim}{j} + \begin{vmatrix} 2 & 1 \\ -4 & 3 \end{vmatrix} \underset{\sim}{k}$$

$$= -2\underset{\sim}{i} - 6\underset{\sim}{j} + 10\underset{\sim}{k}$$

$$\underset{\sim}{v} \times \underset{\sim}{u} = -(\underset{\sim}{u} \times \underset{\sim}{v})$$

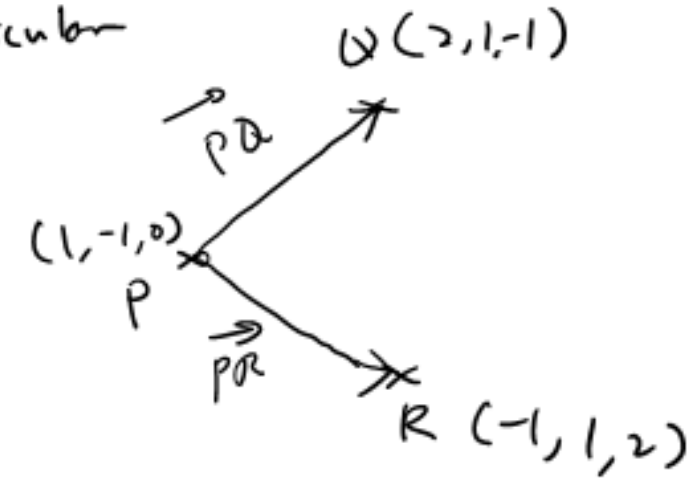
$$= 2\underset{\sim}{i} + 6\underset{\sim}{j} - 10\underset{\sim}{k}$$

Ex: Find $\underset{\sim}{v} \times \underset{\sim}{u}$ by using the determinant formula.

Calculating Cross Product

Example: Find a vector perpendicular to the plane of $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.

We know that $\vec{PQ} \times \vec{PR}$ is perpendicular to both $\vec{PQ}$ and $\vec{PR}$ and hence it must be perpendicular to the plane formed by $\vec{PQ}$ and $\vec{PR}$.



$$\vec{PQ} = \vec{OQ} - \vec{OP} = \langle 2, 1, -1 \rangle - \langle 1, -1, 0 \rangle$$

$$= \langle 1, 2, -1 \rangle$$

$$\vec{PR} = \vec{OR} - \vec{OP} = \langle -1, 1, 2 \rangle - \langle 1, -1, 0 \rangle$$

$$= \langle -2, 2, 2 \rangle$$

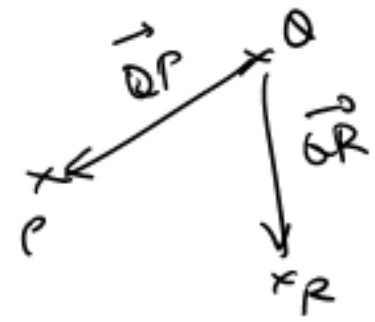
$$\vec{PQ} \times \vec{PR} = \langle 1, 2, -1 \rangle \times \langle -2, 2, 2 \rangle$$

$$= \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & -1 \\ 2 & 2 \end{vmatrix} \underset{\sim}{i} - \begin{vmatrix} 1 & -1 \\ -2 & 2 \end{vmatrix} \underset{\sim}{j} + \begin{vmatrix} 1 & 2 \\ -2 & 2 \end{vmatrix} \underset{\sim}{k}$$

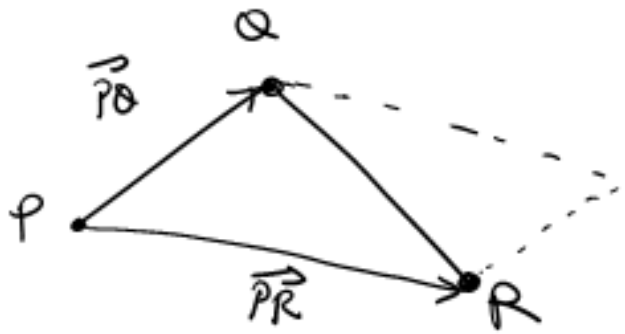
$$= 6\underset{\sim}{i} + 6\underset{\sim}{k}$$

Of course, similarly, you can also use $\vec{PR} \times \vec{PQ}$ or $\vec{QP} \times \vec{QR}$ or $\vec{RP} \times \vec{RQ}$ or their commutative.



Calculating Cross Product

Example: Find the area of the triangle with vertices $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.



Area of triangle formed by $\vec{PQ}$ and $\vec{PR}$

$= \frac{1}{2} \times$ Area of parallelogram formed by $\vec{PQ}$ and $\vec{PR}$

$= \frac{1}{2} \times |\vec{PQ} \times \vec{PR}|$

$= \frac{1}{2} \times \sqrt{6^2 + 6^2} = \frac{1}{2}\sqrt{2(36)} = 3\sqrt{2}.$

Calculating Cross Product

Example: Find a unit vector perpendicular to the plane of $P(1,-1,0)$, $Q(2,1,-1)$, and $R(-1,1,2)$.

We already know that $\vec{PQ} \times \vec{PR}$ is perpendicular to the plane. So we need to find a unit vector $\underset{\sim}{n}$ parallel to the $\vec{PQ} \times \vec{PR}$.

$$\underset{\sim}{n} = \frac{\vec{PQ} \times \vec{PR}}{|\vec{PQ} \times \vec{PR}|} = \frac{6\underset{\sim}{i} + 6\underset{\sim}{k}}{6\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}}\underset{\sim}{i} + \frac{1}{\sqrt{2}}\underset{\sim}{k}$$

Calculating Cross Product

Example: Show that $|\boldsymbol{u} \cdot \boldsymbol{v}|^2 + |\boldsymbol{u} \times \boldsymbol{v}|^2 = |\boldsymbol{u}|^2 |\boldsymbol{v}|^2$.

$$|\underset{\sim}{u} \times \underset{\sim}{v}| = |\underset{\sim}{u}|\,|\underset{\sim}{v}| \sin\theta$$

$$|\underset{\sim}{u} \times \underset{\sim}{v}|^2 = |\underset{\sim}{u}|^2 |\underset{\sim}{v}|^2 \sin^2\theta .$$

$$|\underset{\sim}{u} \times \underset{\sim}{v}|^2 = |\underset{\sim}{u}|^2 |\underset{\sim}{v}|^2 (1 - \cos^2\theta)$$

$$= |\underset{\sim}{u}|^2 |\underset{\sim}{v}|^2 - |\underset{\sim}{u}|^2 |\underset{\sim}{v}|^2 \cos^2\theta$$

$$= |\underset{\sim}{u}|^2 |\underset{\sim}{v}|^2 - |\underset{\sim}{u} \cdot \underset{\sim}{v}|^2$$

$$|\underset{\sim}{u} \cdot \underset{\sim}{v}|^2 + |\underset{\sim}{u} \times \underset{\sim}{v}|^2 = |\underset{\sim}{u}|^2 |\underset{\sim}{v}|^2$$

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = |\underset{\sim}{u}|\,|\underset{\sim}{v}| \cos\theta$$

$$|\underset{\sim}{u} \cdot \underset{\sim}{v}| = |\underset{\sim}{u}|\,|\underset{\sim}{v}|\,|\cos\theta|$$

$$|\underset{\sim}{u} \cdot \underset{\sim}{v}|^2 = |\underset{\sim}{u}|^2 |\underset{\sim}{v}|^2 |\cos^2\theta|$$

Triple Scalar Product

vector · vector $\Rightarrow$ $(\underset{\sim}{u} \times \underset{\sim}{v}) \cdot \underset{\sim}{w}$ should return to a number/scalar.

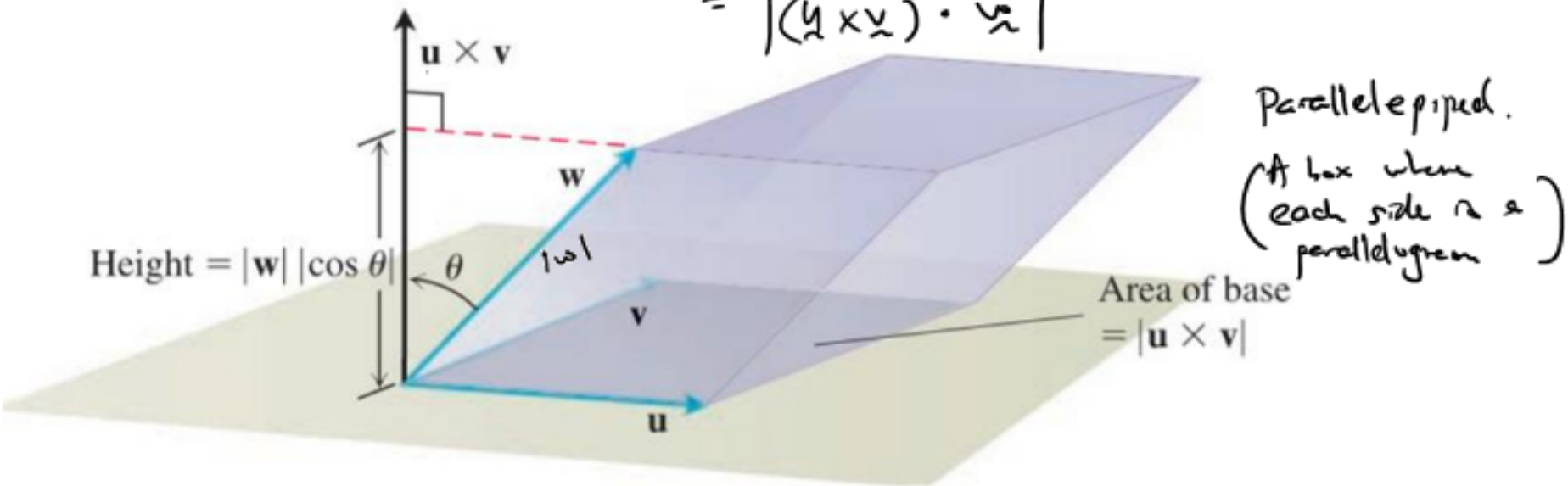
The product $(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w}$ is called the triple scalar product (also known as box product) of $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$.

Volume of parallelepiped = Area of the base × height.

= (Area of parallelogram formed by $\underset{\sim}{u}$ and $\underset{\sim}{v}$) (height)

$= |\underset{\sim}{u} \times \underset{\sim}{v}| \left(|\underset{\sim}{w}| |\cos\theta| \right)$

$= |(\underset{\sim}{u} \times \underset{\sim}{v}) \cdot \underset{\sim}{w}|$



Parallelepiped.
(A box where each side is a parallelogram)

Volume = area of base · height

Properties of Triple Scalar Product

$(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w} = (\boldsymbol{v} \times \boldsymbol{w}) \cdot \boldsymbol{u} = (\boldsymbol{w} \times \boldsymbol{u}) \cdot \boldsymbol{v}$.

As long as the order follow $\underset{\sim}{u} \to \underset{\sim}{v} \to \underset{\sim}{w}$

We also know that the parallelogram formed by $\underset{\sim}{v}$ and $\underset{\sim}{w}$ or $\underset{\sim}{u}$ and $\underset{\sim}{w}$ can also be the base of parallelepiped.

$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ are coplanar iff $(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w} = 0$

$\underset{\sim}{u} \times \underset{\sim}{v}$ is perpendicular to the plane formed by $\underset{\sim}{u}$ and $\underset{\sim}{v}$

So, $\underset{\sim}{u}$, $\underset{\sim}{v}$ and $\underset{\sim}{w}$ are coplanar

$\iff$ $\underset{\sim}{w}$ also lies on the plane formed by $\underset{\sim}{u}$ and $\underset{\sim}{v}$

$\iff$ $\underset{\sim}{w}$ is perpendicular to $\underset{\sim}{u} \times \underset{\sim}{v}$

$\iff$ $(\underset{\sim}{u} \times \underset{\sim}{v}) \cdot \underset{\sim}{w} = 0$.

* If any two vectors of $\underset{\sim}{u}, \underset{\sim}{v}, \underset{\sim}{w}$ are parallel, then the triple scalar product is 0.

* $(\underset{\sim}{u} \times \underset{\sim}{u}) \cdot \underset{\sim}{v} = (\underset{\sim}{u} \times \underset{\sim}{v}) \cdot \underset{\sim}{u} = (\underset{\sim}{u} \times \underset{\sim}{v}) \cdot \underset{\sim}{v} = 0$

Triple Scalar Product as Determinant

Let $\boldsymbol{u} = u_1\boldsymbol{i} + u_2\boldsymbol{j} + u_3\boldsymbol{k}$, $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$ and $\boldsymbol{w} = w_1\boldsymbol{i} + w_2\boldsymbol{j} + w_3\boldsymbol{k}$.

$$(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w} = \left(\begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} \right) \cdot \left(w_1\underset{\sim}{i} + w_2\underset{\sim}{j} + w_3\underset{\sim}{k} \right)$$

$$= \left(\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \underset{\sim}{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \underset{\sim}{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \underset{\sim}{k} \right) \cdot \left(w_1\underset{\sim}{i} + w_2\underset{\sim}{j} + w_3\underset{\sim}{k} \right)$$

$$= w_1 \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - w_2 \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + w_3 \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

$$= \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

← determinant formula for triple scalar product.

Calculating Triple Scalar Product

Example: Find the volume of the parallelepiped determined by $\boldsymbol{u} = \boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$, $\boldsymbol{v} = -2\boldsymbol{i} + 3\boldsymbol{k}$ and $\boldsymbol{w} = 7\boldsymbol{j} - 4\boldsymbol{k}$.

$$\text{Volume} = \left| (\underset{\sim}{u} \times \underset{\sim}{v}) \cdot \underset{\sim}{w} \right|$$

Ex: Try $(\underset{\sim}{u} \times \underset{\sim}{v}) \cdot \underset{\sim}{v}$

$$(\underset{\sim}{u} \times \underset{\sim}{v}) \cdot \underset{\sim}{w} = \begin{vmatrix} 1 & 2 & -1 \\ -2 & 0 & 3 \\ 0 & 7 & -4 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 0 & 3 \\ 7 & -4 \end{vmatrix} - (-2) \begin{vmatrix} 2 & -1 \\ 7 & -4 \end{vmatrix} + 0$$

$$= -21 + 2(-1) = -23$$

$$\text{Volume} = |-23| = 23 \text{ unit}^3.$$

Calculating Triple Scalar Product

Example: Show that $\boldsymbol{u} = \boldsymbol{i} + \boldsymbol{j} + \boldsymbol{k}$, $\boldsymbol{v} = \boldsymbol{i} + 2\boldsymbol{j} + 3\boldsymbol{k}$ and $\boldsymbol{w} = 2\boldsymbol{i} + 3\boldsymbol{j} + 4\boldsymbol{k}$ are coplanar.

$$(\underset{\sim}{u} \times \underset{\sim}{v}) \cdot \underset{\sim}{w} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{vmatrix}$$

$$= 1\begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} - \begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix}$$

$$= -1 + 2 - 1 = 0$$

So, $\underset{\sim}{u}$, $\underset{\sim}{v}$ and $\underset{\sim}{w}$ are coplanar.